

# RobinReal

Hybrid Listing Search & Relevance Ranking

*Bridging strict constraints and human preferences  
with LLMs and Vision–Language Models*

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# The Core Challenge

Users mix **strict rules** with **subjective wishes**:

*“Ich suche eine **2,5-Zimmer-Wohnung** in **Zürich**, max **2’800 CHF**, mit **Balkon**. Wichtig sind mir eine *moderne Küche* und ein *lebendiges, aber nicht zu lautes Quartier*.”*

## ■ Hard constraint

→ SQL WHERE filter

## ■ Soft preference

→ Ranking signal (needs vision)

Strict-only  $\Rightarrow$  zero results. Soft-only  $\Rightarrow$  ignores budget.

# Unlocking Subjective Signals with VLMs

Text alone cannot answer “*bright*”, “*modern*”, “*nice view*”.

We turn listing **images** into structured ranking data with **Qwen2-VL-7B**.

## 1. Ground-truth flags

`has_greenery_visible: true`  
Verifies what text claims.

## 2. Appeal scores (1–5)

Brightness, Modernity,  
Spaciousness.  
Fuel for soft-ranking.

## 3. Visual summaries

Natural-language descriptions.  
Semantic search on *actual*  
appearance.

# Case Study: Fine-Grained Material Recognition

Listing 69afff8a0ad89f01c7f09fd2 — Soft labels all 5/5.

What matters is **what the VLM *actually* extracted**:



*“Plank flooring” / “Wooden flooring”*



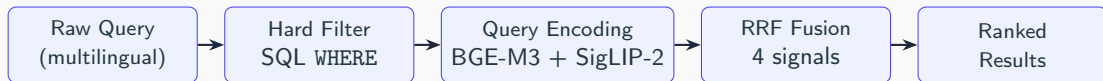
*“electric skylight”*

**Extracted** reason\_phrases\_en

granite countertops • plank flooring • wooden flooring • electric skylight

The model sees materials and specific fixtures — not just “a kitchen”.

# Multi-Modal Retrieval Pipeline



**BGE Dense**  
semantic sim.

**BGE Sparse**  
token overlap

**SigLIP-2**  
text ↔ image

**BM25**  
multilingual keyword

**Reciprocal Rank Fusion** ( $k=60$ ):  $\text{score}[l] = \sum_i \frac{1}{60 + \text{rank}_i(l)}$

# Why RRF + Diversity Injection?

## The problem with single signals

- Dense embeddings miss rare keywords
- Keyword search misses paraphrase
- One image can dominate the top-k

## Our fix

- **RRF** combines 4 rank lists — no single signal monopolises
- **BM25 injection** fills embedding blind spots for edge-case terms

*Robustness comes from disagreement between signals, not from one “best” model.*

# Handling “Impossible” Queries

*“Modern apartment in Basel, cheap but central, quiet with nice views”*

These traits are inherently opposed. Instead of guessing, we return **three ranked clusters**:

## Strategy A: Price

2.5-Zi Kleinbasel

**CHF 1,350/mo**

- + central Kleinbasel
- street noise
- not “modern”

## Strategy B: Balance

3.5-Zi Gundeldingen

**CHF 1,950/mo**

- + modern new build
- + 5th floor views
- + good transit

## Strategy C: Lifestyle

3.5-Zi St. Alban

**CHF 2,800/mo**

- + premium central
- + Rhine views
- expensive

We hand the trade-off back to the user — transparently.

# From Concept to Deployment: MCP Server

The pipeline is exposed as a **Model Context Protocol** tool (`search_listings`), callable directly by agents (e.g. Cursor, Claude).

## ↑ Rule-Based Extraction

- Regex patterns + Swiss gazetteer
- AMTOVZ\_CSV\_LV95.csv (15k+ locations)
- Covers ~90% of common queries
- Fast • Free

## ⊗ LLM-Powered Extraction

- Claude API with structured tools
- Multilingual, contradictory queries
- Nuanced soft-preference extraction
- Flexible • Edge-case resilient

**Stack:** FastMCP • SQLite + FastAPI • Docker • Cloudflare Tunnel



# Takeaways

1. **Separate intent:** hard filters in SQL, soft preferences in ranking.
2. **VLMs unlock pixels:** Qwen2-VL turns images into scores *and* fine-grained phrases (“granite countertops”, “electric skylight”).
3. **Fuse, don’t pick:** RRF over BGE-dense, BGE-sparse, SigLIP-2, BM25.
4. **Trade-offs are the product:** surface alternative clusters instead of one answer.
5. **Ship it:** deployed as an MCP tool, usable by any LLM agent.

Thank you.

Questions?